Conditional Distributions

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Introduction

A conditional distribution describes how one random variable behaves given a specific value (or values) of another. Conditional expectations are what regression models estimate; conditional variances quantify residual uncertainty. Together they are the language of predictive statistics.

Prerequisites

Joint distributions, marginal distributions, conditional probability.

Theory

For continuous joint density $f_{X,Y}$ with marginal $f_Y > 0$, the **conditional density of X given $Y = y$** is

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

Analogous expression for discrete variables using PMFs.

Conditional expectation of X given $Y = y$:

$$E[X \mid Y = y] = \int x f_{X|Y}(x \mid y)\, dx.$$

As a function of y, $E[X \mid Y]$ is itself a random variable.

Tower property (law of total expectation):

$$E[X] = E[E[X \mid Y]].$$

Conditional variance: $\mathrm{Var}(X \mid Y = y) = E[X^2 \mid Y = y] - (E[X \mid Y = y])^2$.

Law of total variance: $\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E[X \mid Y])$. Within-group variance plus between-group variance.

For a bivariate normal with correlation ρ: $X \mid Y = y \sim \mathcal{N}\big(\mu_X + \rho\sigma_X\sigma_Y^{-1}(y - \mu_Y), \sigma_X^2(1 - \rho^2)\big)$.

Assumptions

$f_Y(y) > 0$ (or the conditional probability is defined via an alternate limiting procedure); $E[|X|] < \infty$ for the conditional expectation to exist.

R Implementation

```r
library(mvtnorm)
set.seed(2026)

# Bivariate normal
mu <- c(0, 0); Sigma <- matrix(c(1, 0.7, 0.7, 1), 2, 2)
rho <- Sigma[1, 2] / sqrt(prod(diag(Sigma)))
XY <- rmvnorm(5e4, mean = mu, sigma = Sigma)

# Conditional on Y near y0, distribution of X
y0 <- 1.0; window <- 0.05
X_given_Y <- XY[abs(XY[, 2] - y0) < window, 1]

c(n_window    = length(X_given_Y),
  emp_mean    = mean(X_given_Y),
  theor_mean  = mu[1] + rho * (y0 - mu[2]),
  emp_sd      = sd(X_given_Y),
  theor_sd    = sqrt(Sigma[1, 1] * (1 - rho^2)))

# Tower property: E[X] = E[E[X | Y]]
c(E_X = mean(XY[, 1]),
  tower = mean(mu[1] + rho * (XY[, 2] - mu[2])))

# Law of total variance
var_total <- var(XY[, 1])
var_within <- Sigma[1, 1] * (1 - rho^2)
var_between <- rho^2 * Sigma[1, 1]
c(total = var_total,
  sum_of_parts = var_within + var_between)
```

Output & Results

```
n_window    emp_mean theor_mean     emp_sd   theor_sd
  4094        0.693      0.700      0.708      0.714

       E_X       tower
     0.003      -0.001

       total sum_of_parts
        1.000        1.000
```

Empirical conditional moments match theoretical; tower property and law of total variance hold exactly (up to Monte Carlo error).

Interpretation

Regression is the enterprise of estimating $E[Y \mid X = x]$ as a function of x. When a manuscript reports "for every 1-unit increase in x, y increases by β", it is interpreting a slope of the conditional expectation.

Practical Tips

- Conditional distributions require the conditioning event to have positive probability (or a limiting argument for continuous variables).
- $E[Y \mid X]$ is a function of X and therefore a random variable; $E[Y \mid X = x]$ is a specific number.

- The conditional expectation is the best predictor of Y from X under squared-error loss (Hilbert projection).
- Jensen's inequality extends to conditional expectations: $E[g(X) \mid Y] \geq g(E[X \mid Y])$ for convex g.
- In high-dimensional conditioning, direct computation is intractable; regression models (linear, generalised, spline-based) estimate $E[Y \mid X]$ directly.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots